Topic: Systems of linear inequalities

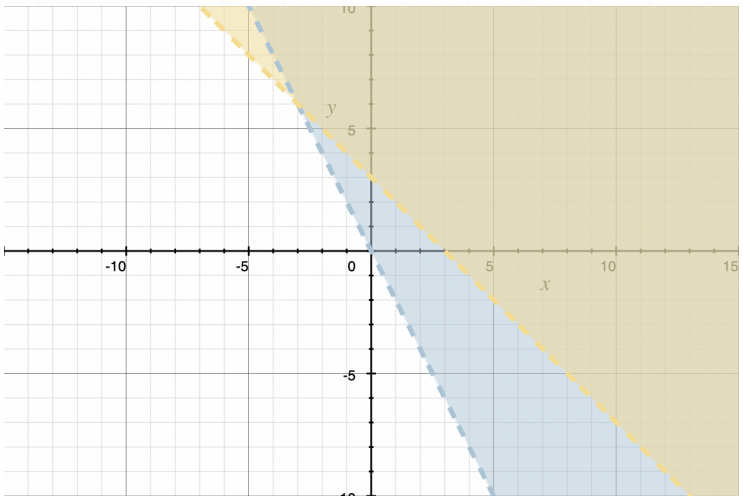
Question: Graph the solution of the system of linear inequalities.

$$2x + y > 0$$

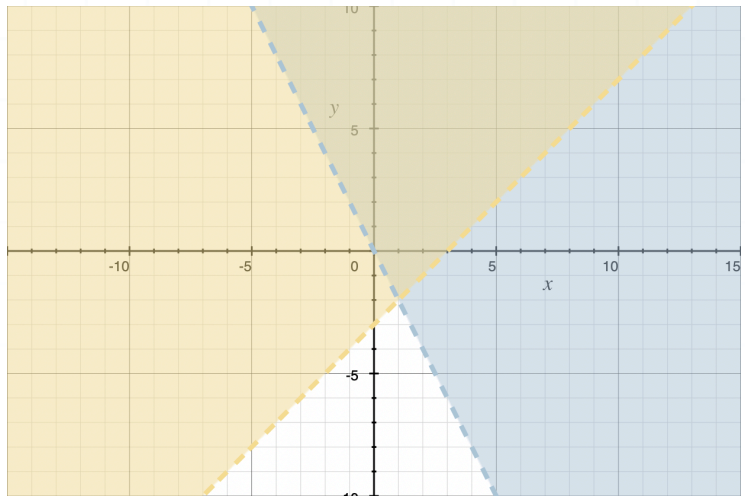
$$x - y < 3$$

Answer choices:

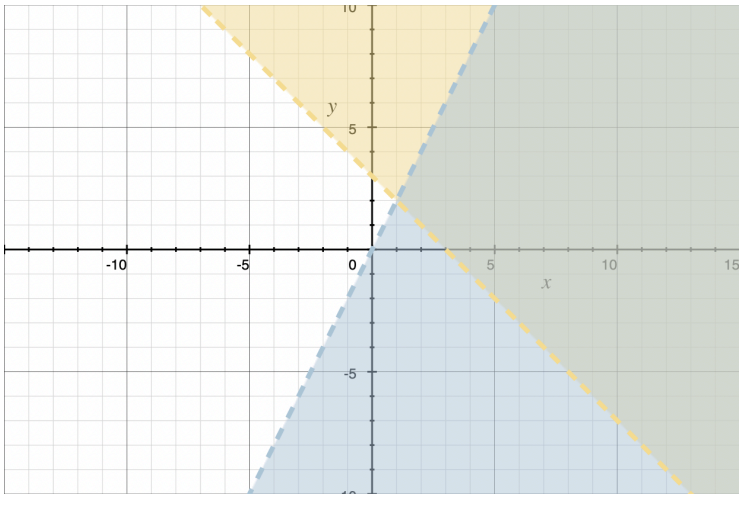
A



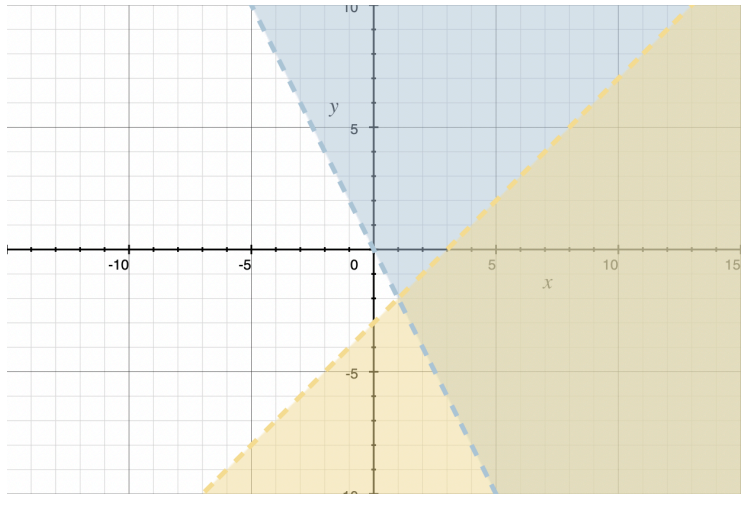
B



C



D



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Solution: B

We need to find the boundary lines of each inequality by graphing their corresponding equations. We'll rewrite the corresponding equations in slope-intercept form.

$$2x + y = 0$$

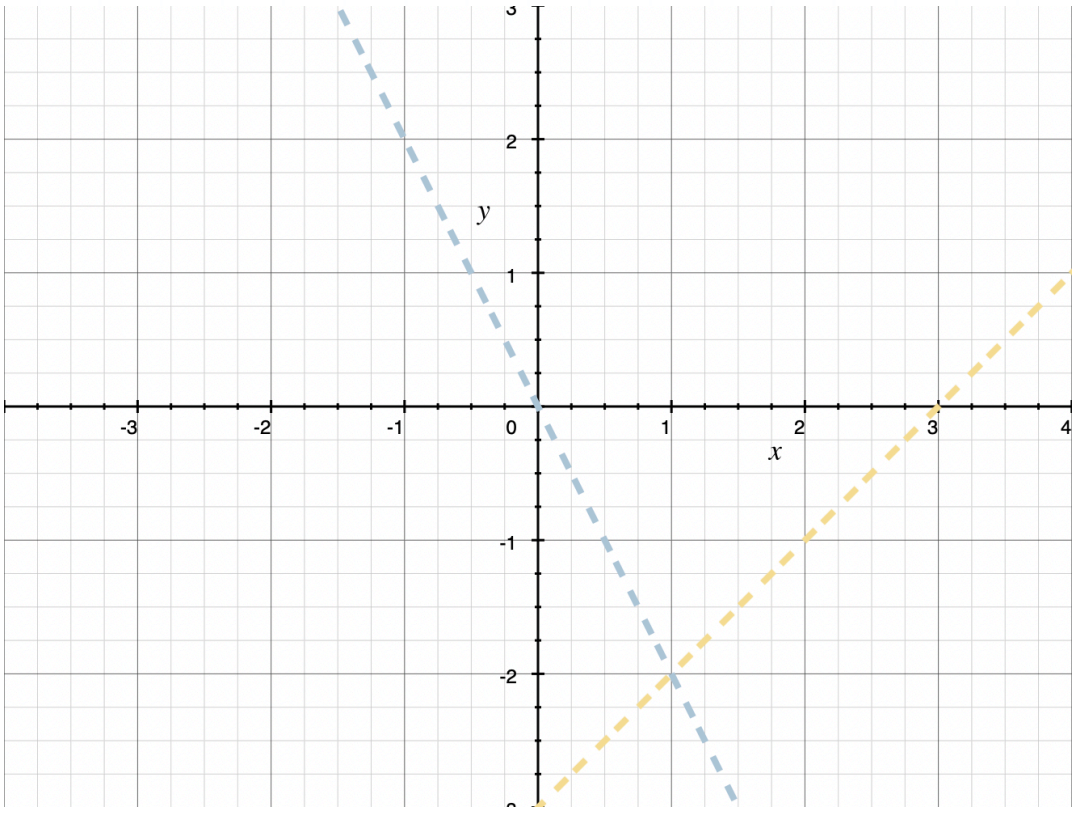
$$y = -2x$$

and

$$x - y = 3$$

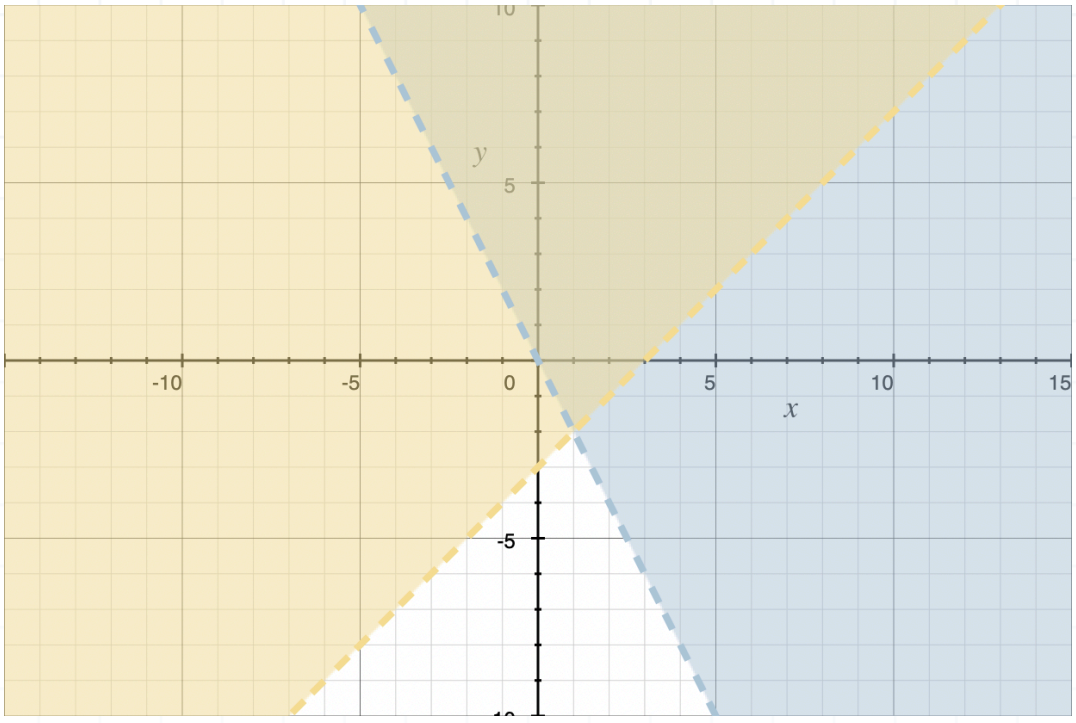
$$y = x - 3$$

Because the inequalities are $<$ and $>$, the lines should be dashed.



We need to shade above $y = -2x$ and below $y = x - 3$, and the solution to the system of inequalities is

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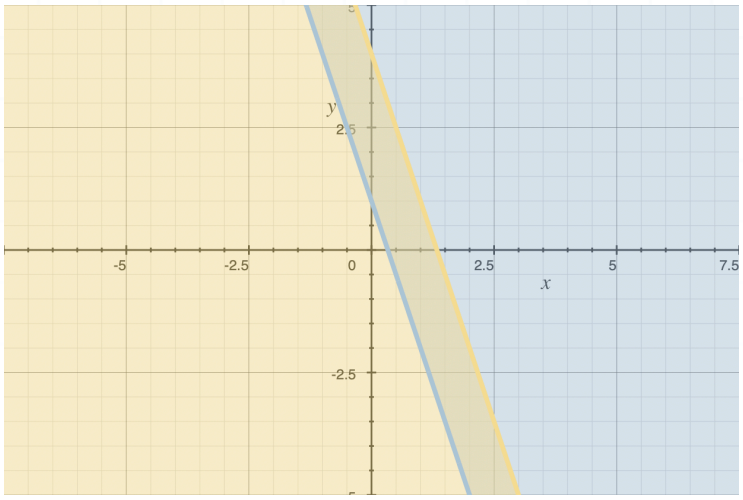
Question: Graph the solution of the system of linear inequalities.

$$3x + y \geq 1$$

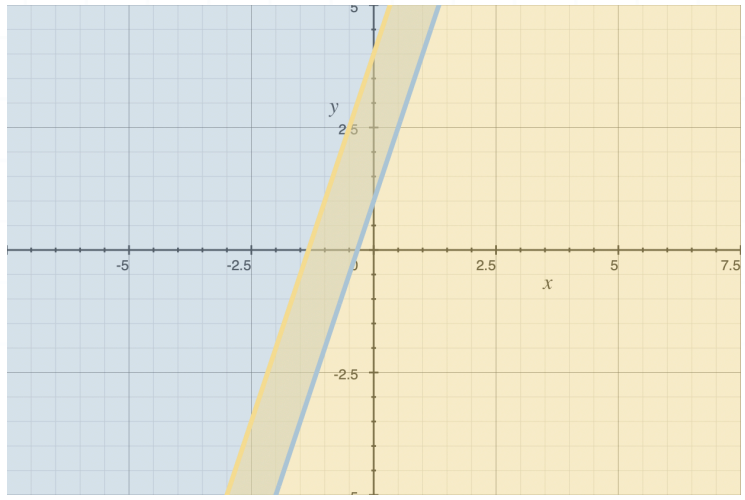
$$3x + y \leq 4$$

Answer choices:

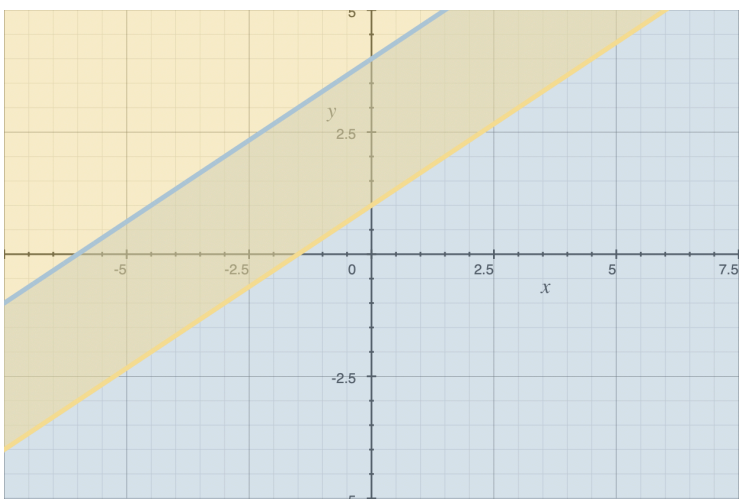
A



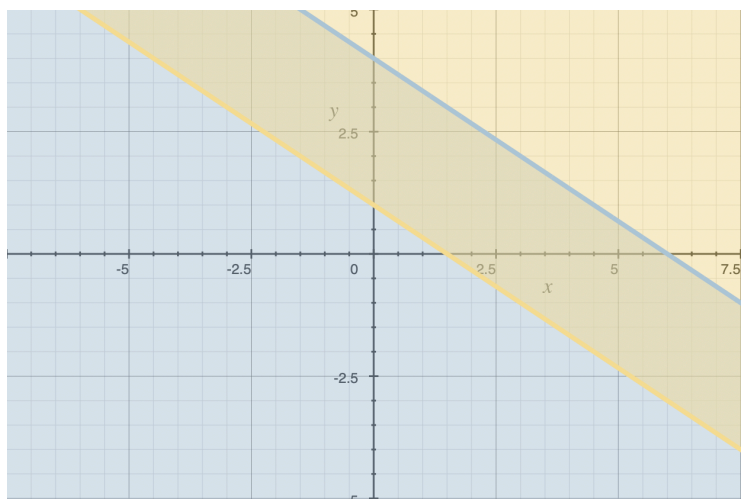
B



C



D



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Solution: A

We need to find the boundary lines of each inequality by graphing their corresponding equations. We'll rewrite the corresponding equations in slope-intercept form.

$$3x + y = 1$$

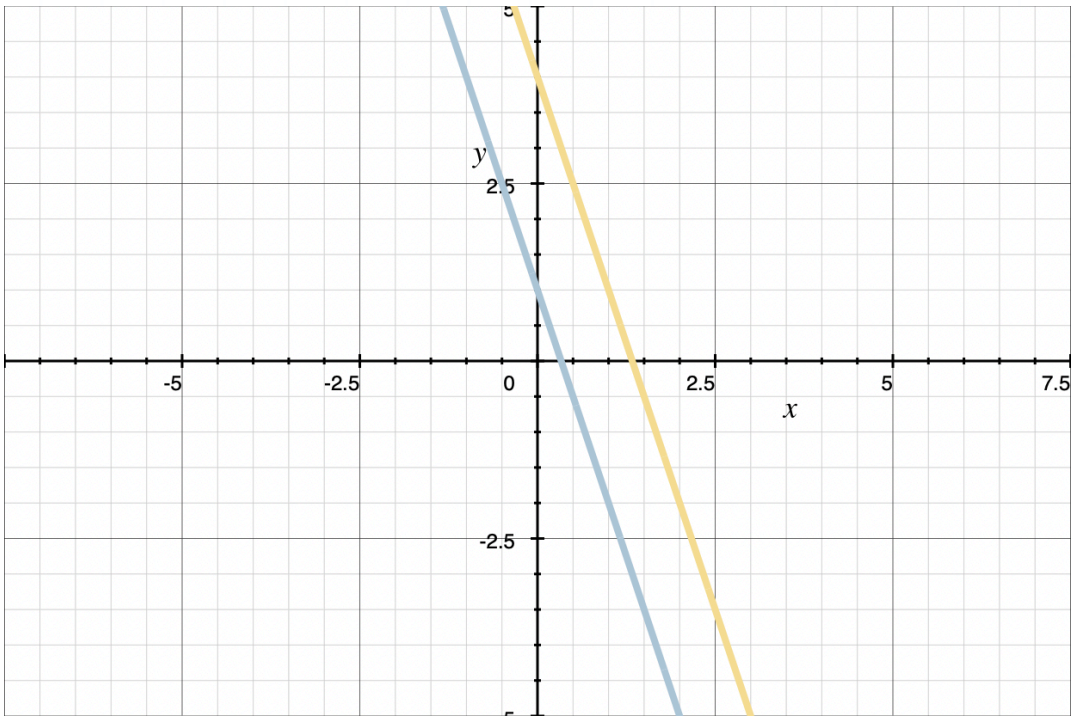
$$y = -3x + 1$$

and

$$3x + y = 4$$

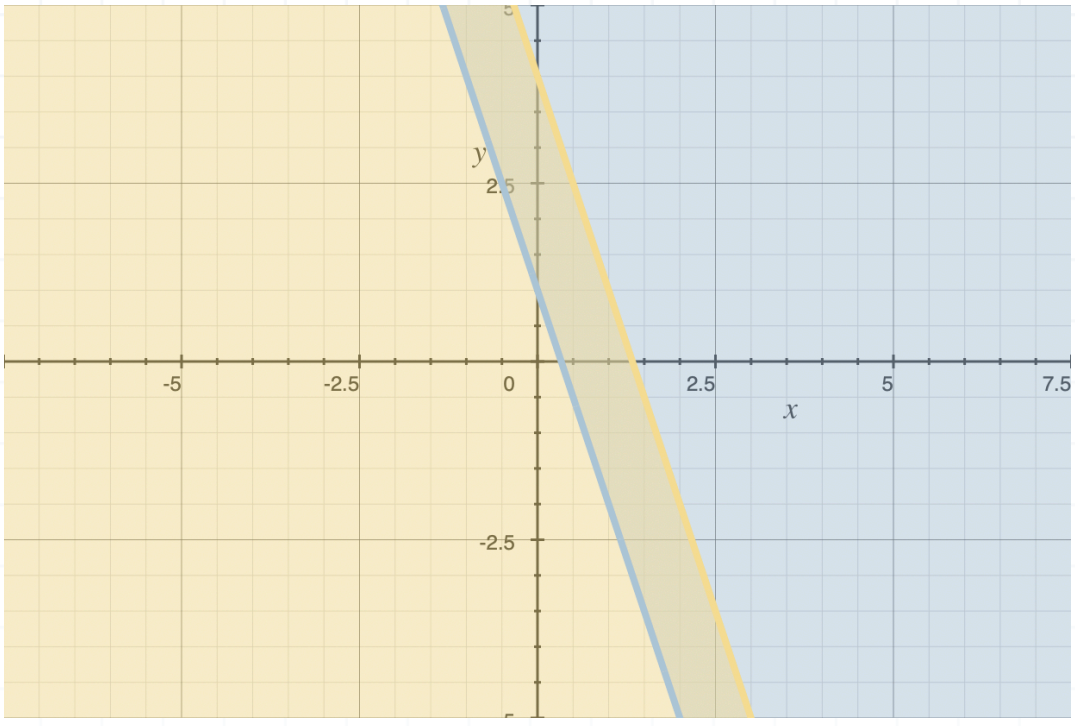
$$y = -3x + 4$$

Because the inequalities are $\leq$ and $\geq$, the lines should be solid.



We need to shade above $y = -3x + 1$ and below $y = -3x + 4$, and the solution to the system of inequalities is

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Topic: Systems of linear inequalities

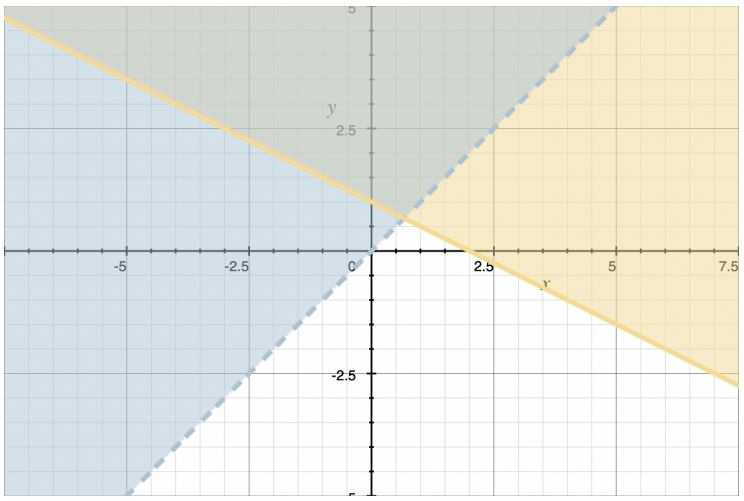
Question: Graph the solution of the system of linear inequalities.

$$x > y$$

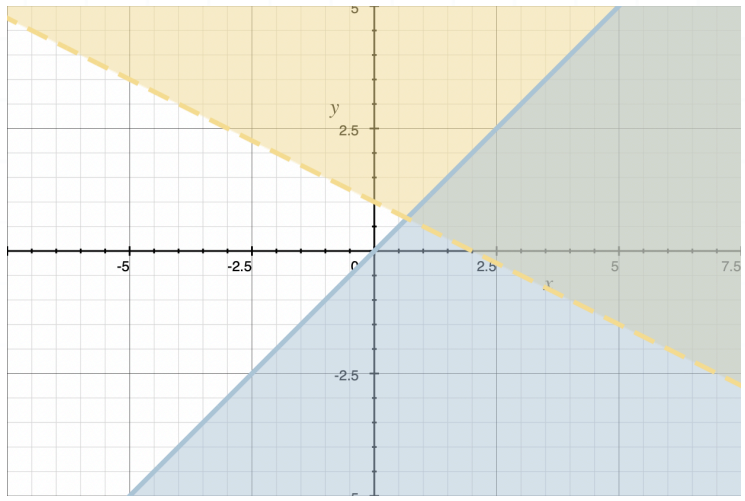
$$x + 2y \geq 2$$

Answer choices:

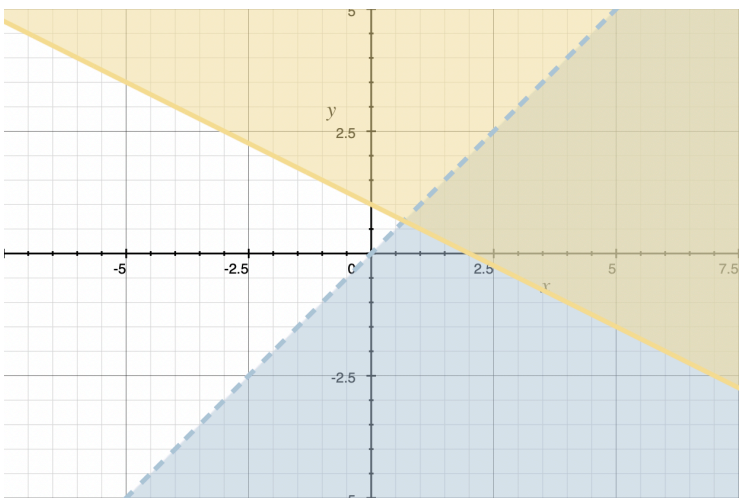
A



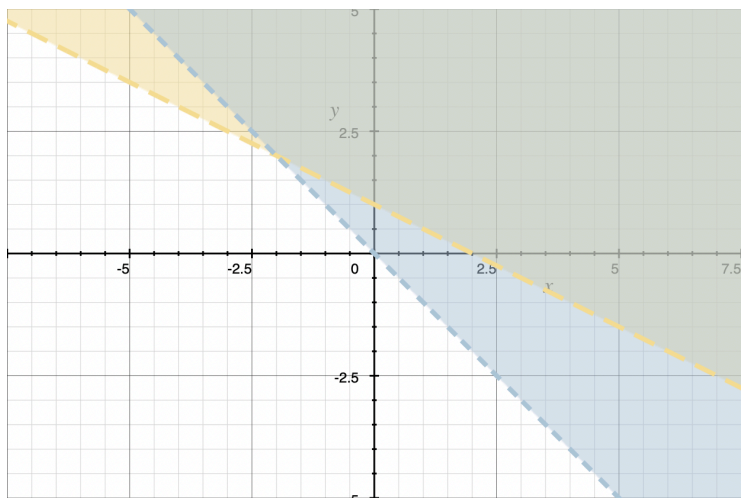
B



C



D



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Solution: C

We need to find the boundary lines of each inequality by graphing their corresponding equations. We'll rewrite the corresponding equations in slope-intercept form.

$$x = y$$

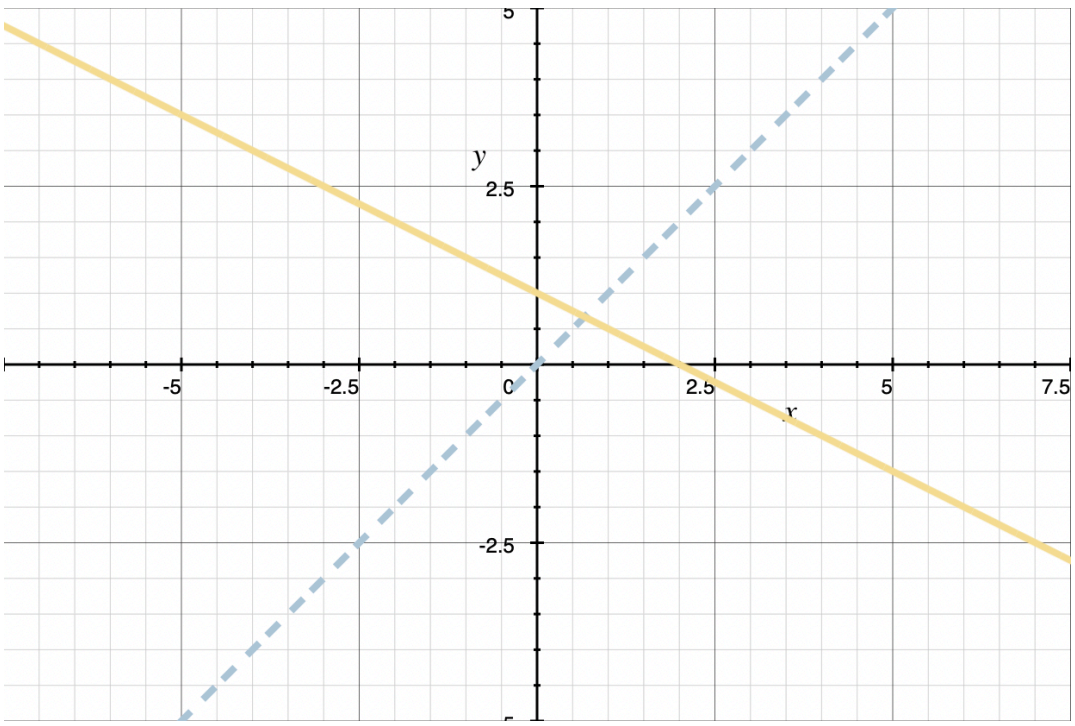
$$y = x$$

and

$$x + 2y = 2$$

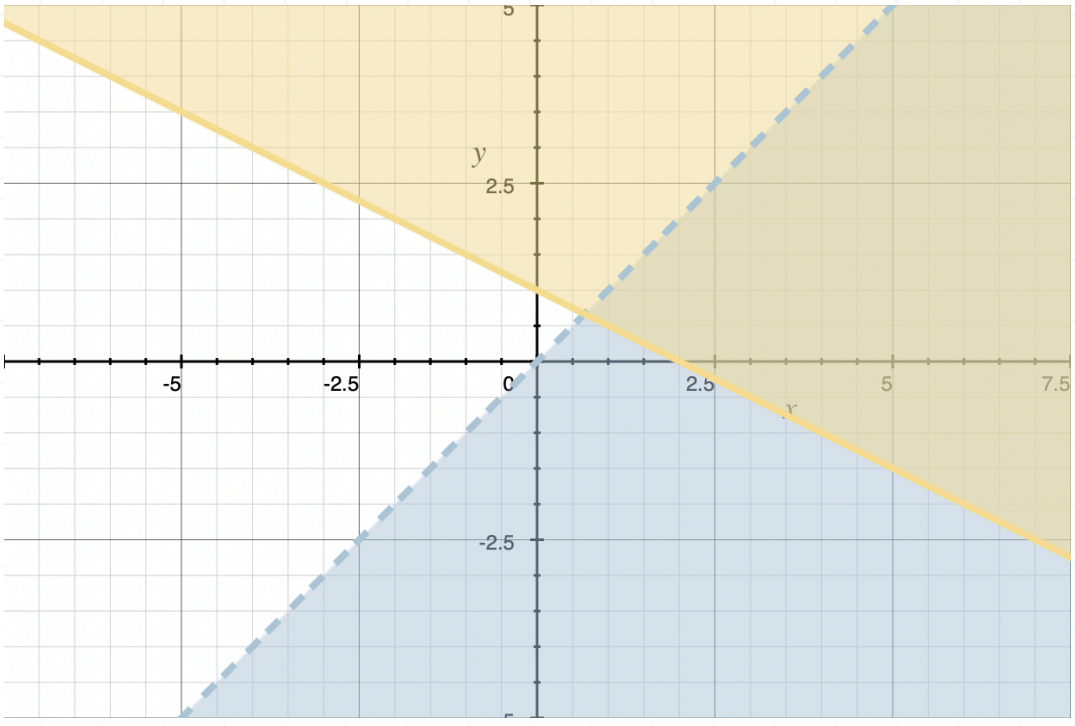
$$y = -\frac{1}{2}x + 1$$

Because the inequalities are $\geq$ and $>$, one line will be dashed and the other will be solid.



We need to shade below $y = x$ and above $y = -(1/2)x + 1$, and the solution to the system of inequalities is

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